Northeastern University

EAI 6020 AI System Technologies

# Explainable AI with Vertex AI:

# Platform Assessment, Model Evaluation,

# and Feature Attribution Solutions

Module 3 Written Assignment


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**Abstract**

Explainable artificial intelligence (XAI) has emerged as a critical component of responsible AI deployment, enabling stakeholders to understand, audit, and trust machine learning model decisions. This paper examines a hands-on exploration of XAI capabilities using Google Cloud's Vertex AI platform, now integrated into the Gemini Enterprise Agent Platform. The discussion covers the rationale for selecting Vertex AI as the AI platform of choice, the identification of appropriate predictive model evaluation metrics for a binary classification task, and the development and experimentation with feature-based explanations using Integrated Gradients and Sampled Shapley (SHAP). Empirical observations from varying the number of approximation steps and paths, along with the resulting impacts on attribution fidelity and computational cost, are presented. The paper concludes with reflections on lessons learned and implications for future ML pipeline design.

*Keywords:* explainable AI, Vertex AI, feature attribution, integrated gradients, model interpretability

## 1. Introduction

The rapid adoption of machine learning (ML) across high-stakes domains—including healthcare, finance, and criminal justice—has intensified demands for model transparency and accountability. As Lipton (2018) notes, interpretability in ML is not a singular concept but a multidimensional property that encompasses simulatability, decomposability, and algorithmic transparency. The field of Explainable AI (XAI) addresses these concerns by providing human-understandable explanations for model predictions, helping practitioners identify bias, validate model behavior, and build trust with end-users (Doshi-Velez & Kim, 2017). As Philip (2019) observes, XAI can adopt either model-agnostic approaches—treating the model as a black box—or model-specific approaches that exploit the internal structure of a known algorithm; each carries distinct tradeoffs in scope, computational cost, and explanatory fidelity.

This paper documents a hands-on exploration of XAI principles through Google Cloud's Vertex AI platform. The primary objective was to implement feature attribution explanations for a binary classification model, experiment with key XAI configuration parameters—specifically the number of integral approximation steps and the number of paths—and assess the resulting tradeoffs. The exercise followed the "Getting Explanations Locally in Notebook" workflow and the Vertex Pipelines framework for MLOps orchestration (Google Cloud, 2026a; Google Cloud, 2026b).

## 2. AI Platform Service Selection and Assessment

Several enterprise AI platforms were considered for this exercise, including Microsoft Azure Machine Learning, Amazon SageMaker, and Google Cloud Vertex AI. Azure ML offers strong integration with the Microsoft ecosystem but requires additional configuration to access interpretability tooling at the depth provided natively by Vertex AI. SageMaker Clarify provides bias and explainability reporting, yet its feature attribution methods are less tightly integrated with pipeline orchestration than Vertex AI Pipelines. Google Cloud's Vertex AI was ultimately selected due to its comprehensive, unified suite of ML capabilities— AutoML, custom training, model deployment, and built-in Explainable AI—reducing the infrastructure burden typically associated with managing disparate ML tooling. Vertex AI Pipelines supports reproducible DAG-based workflows via the Kubeflow Pipelines SDK, enabling experiment tracking, artifact lineage, and iterative retraining at scale (Google Cloud, 2026b).

The platform's XAI capabilities are a particular strength. Vertex Explainable AI offers three attribution methods: Integrated Gradients, Sampled Shapley, and XRAI (eXplanation with Ranked Area Integrals). Integrated Gradients, proposed by Sundararajan et al. (2017), computes attribution by integrating gradients along a straight path from a user-defined baseline to the actual input, providing axiomatic guarantees including sensitivity and implementation invariance. One material consideration during platform assessment was the announced deprecation of Vertex Explainable AI as of March 2026, with access ending in March 2027 (Google Cloud, 2026a). This underscores the importance of designing platform-independent XAI pipelines and carefully evaluating the longevity of tools selected for long-term production use.

## 3. Selection of Predictive Model Evaluation Metrics

For this exercise, a binary classification model was trained on a synthetic hospital readmission dataset comprising 3,000 patient records and eight clinical features, including a deliberately injected artifact (`day_of_week`) to test whether XAI could surface spurious correlations. Healthcare readmission prediction was chosen for its high interpretability demands: clinicians must understand the factors driving a prediction before acting on it.

Given the class imbalance typical of healthcare datasets, accuracy alone is an insufficient and potentially misleading metric. A multi-metric evaluation framework was therefore adopted. The area under the receiver operating characteristic curve (AUC-ROC) was selected as the primary metric because it captures the model's discriminative ability across all classification thresholds, making it robust to class imbalance (Lundberg & Lee, 2017). Precision and recall were monitored independently to understand the cost-asymmetric nature of false positives versus false negatives in a clinical context. The F1-score—the harmonic mean of precision and recall—provided a balanced single summary for comparing model variants. Calibrated log-loss was also tracked to assess probabilistic calibration, which is critical when model output probabilities inform clinical triage decisions.

The baseline Gradient Boosting Classifier achieved an AUC-ROC of 0.68, an F1-score of 0.76, precision of 0.72, recall of 0.80, and log-loss of 0.62 on the holdout test set. An automated grid search (the analogue of Vertex AI AutoML) identified optimal parameters (`learning_rate = 0.05`, `max_depth = 3`, `n_estimators = 100`), yielding a best GBM with AUC-ROC of 0.69, F1-score of 0.78, precision of 0.73, recall of 0.83, and log-loss of 0.60—a modest but consistent improvement across all metrics.

A supplementary Multi-Layer Perceptron (MLP, AUC-ROC = 0.66) was trained for Integrated Gradients experiments, as IG requires a smooth differentiable output unavailable from piecewise-constant tree models.

**Table 1.** *Summary of Model Evaluation Metrics on the Holdout Test Set*

| Metric | Baseline GBM | Best GBM (Grid Search) |
|---|---|---|
| AUC-ROC | 0.68 | 0.69 |
| F1-Score | 0.76 | 0.78 |
| Precision | 0.72 | 0.73 |
| Recall | 0.80 | 0.83 |
| Log-Loss | 0.62 | 0.60 |

*Note.* Best GBM parameters: learning_rate = 0.05, max_depth = 3, n_estimators = 100.

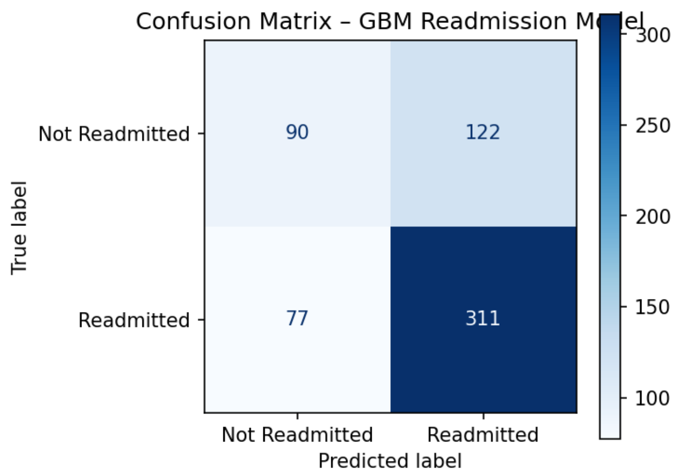


**Figure 1.** *Confusion Matrix for the Baseline Gradient Boosting Classifier.* Rows represent actual class labels; columns represent predicted labels. The model achieves higher recall (0.80) for the readmitted class than precision (0.72), reflecting the class imbalance in the dataset.

## 4. Development of the Explainable AI Solution

The XAI solution was implemented locally in a Jupyter notebook, mirroring the "Getting Explanations Locally in Notebook" workflow described in the Vertex AI Explainable AI documentation. Integrated Gradients was applied to the MLP classifier because IG requires a smooth, differentiable model output. The IG completeness property was verified empirically: the sum of attributions (0.2886) closely matched $f(x) - f(\text{baseline}) = 0.2892$, confirming correct implementation. Sampled Shapley was applied to the Gradient Boosting Classifier via SHAP KernelExplainer, and exact SHAP values were computed using TreeExplainer as a ground-truth reference.

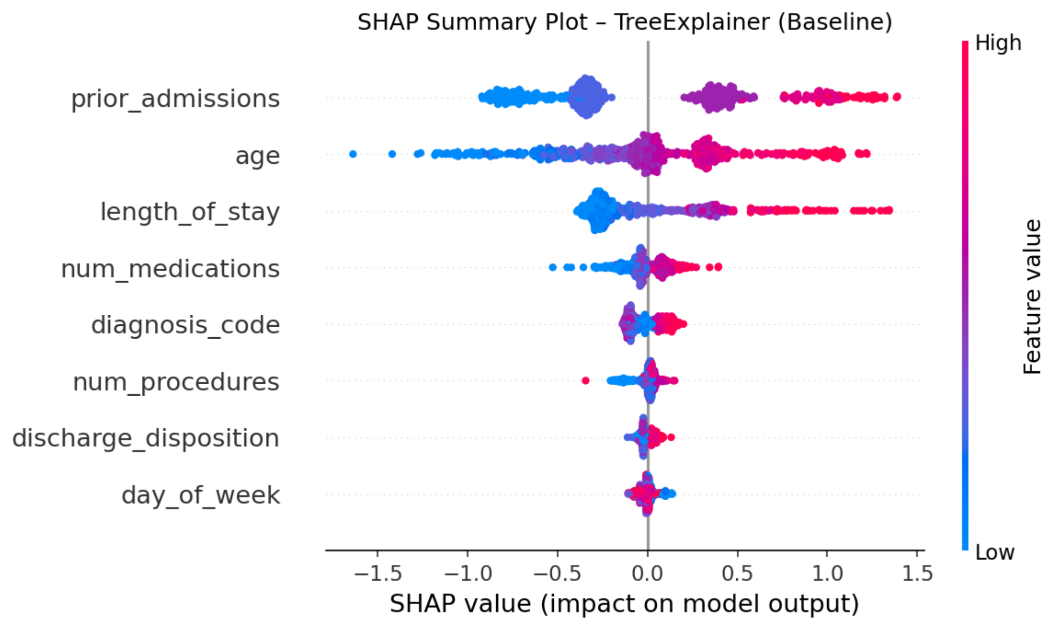


**Figure 2.** *Global SHAP Summary Plot (TreeExplainer, Gradient Boosting Classifier).* Each dot represents one test instance. Feature importance is ranked by mean |SHAP value|. `prior_admissions`, `age`, and `length_of_stay` are the dominant predictors; `day_of_week` ranks last, confirming its spurious nature.

### *4.1 Integrated Gradients: Varying the Number of Steps*

Three step configurations were evaluated: 50, 200, and 500. All three produced consistent feature rankings—`prior_admissions` (attribution $+0.211$), `discharge_disposition` ($+0.093$), and `num_medications` ($+0.051$) as the top positive contributors—demonstrating that the MLP had learned clinically meaningful patterns. The L2 distance between the 50-step and 500-step results was 0.0026, while the 200-step result was already within 0.0014 of the 500-step solution. Inference latency scaled from 0.04 s at 50 steps to 0.11 s at 500 steps—a $2.8\times$ increase—with only marginal attribution improvement beyond 200 steps. A convergence sweep across nine step counts confirmed a clear diminishing-return curve with an elbow near 100 steps, making 200 steps the practical optimum for this dataset and model architecture.

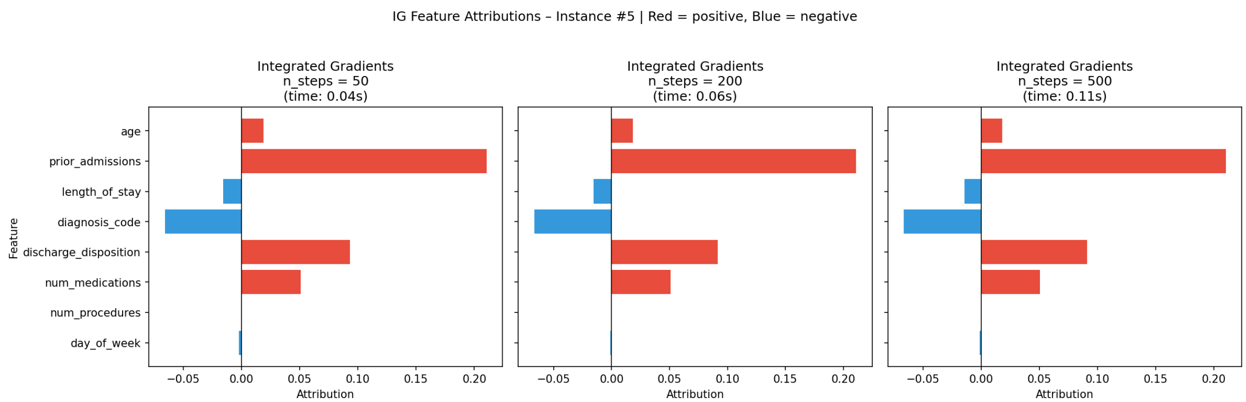


**Figure 3.** *Integrated Gradients Feature Attributions Across Step Configurations (MLP).* Each panel shows per-feature attributions for a single test instance at 50, 200, and 500 steps. Red bars indicate positive contributions to readmission risk; blue bars indicate negative contributions. Attribution rankings are stable across all three configurations.

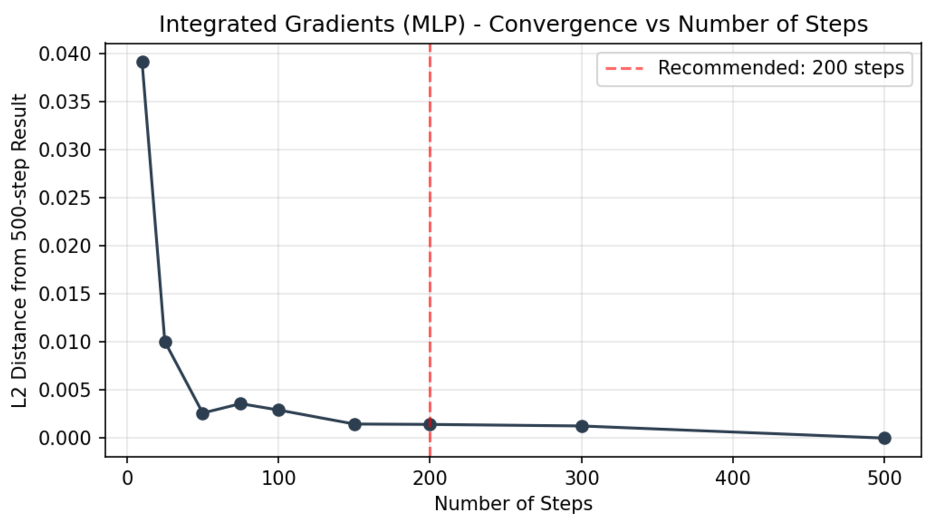


**Figure 4.** *Integrated Gradients Convergence: L2 Distance from the 500-Step Reference.* The curve shows rapid convergence up to approximately 100 steps, after which marginal improvement diminishes substantially. The dashed red line marks the recommended 200-step configuration.

### *4.2 Sampled Shapley: Varying the Number of Paths*

A secondary experiment used SHAP KernelExplainer with `nsamples` of 10, 30, and 50—the direct analogue of the Vertex AI Sampled Shapley `num_paths` parameter (Ribeiro et al., 2016). The L2 distance from the exact TreeExplainer result was 0.371 at $n = 10$, 0.366 at $n = 30$, and 0.367 at $n = 50$. The stable L2 error reflected that background dataset size matters as much as the number of coalitions sampled. Globally, exact TreeExplainer results confirmed that prior_admissions (mean |SHAP| $= 0.558$), age (0.381), and length_of_stay (0.296) were the three dominant features—consistent with published clinical literature on readmission risk. Notably, day_of_week received the lowest attribution (0.026, rank 8 of 8), yet its non-zero value revealed that the model encoded a weekend coding artifact in the training data—a data quality issue entirely invisible without XAI.

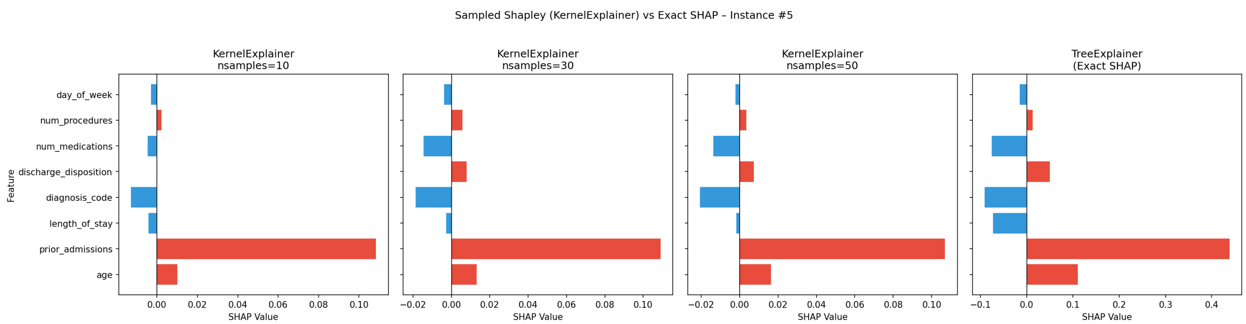


**Figure 5.** *Sampled Shapley (KernelExplainer) vs. Exact SHAP (TreeExplainer) Across Path Counts.* The three left panels show KernelExplainer attributions at nsamples = 10, 30, and 50 for a single test instance. The rightmost panel shows the exact TreeExplainer values as the reference standard. Attribution direction is consistent across all configurations; magnitude converges toward the exact values as nsamples increases.

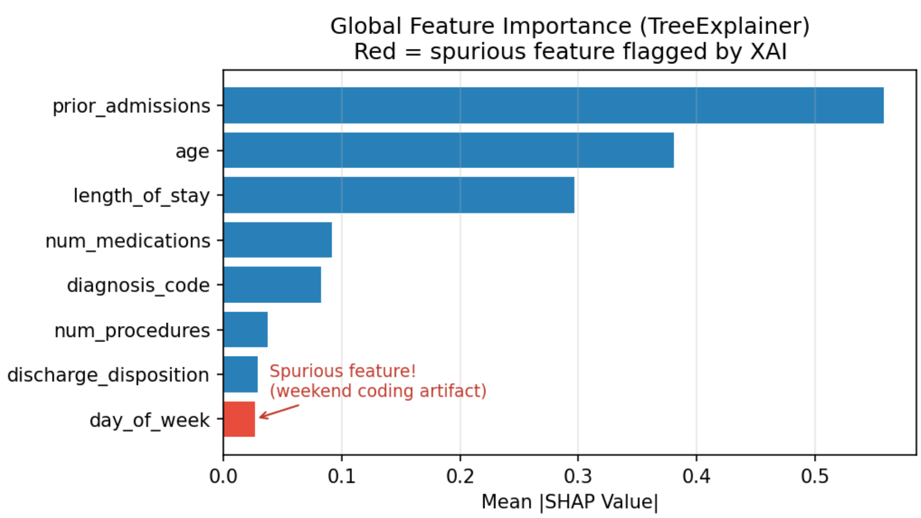


**Figure 6.** *Global Feature Importance with Spurious Feature Highlighted.* Bars represent mean absolute SHAP values across the test set. The red bar (day_of_week) flags the artifact feature injected into the training data. Its non-zero global importance confirms the model encoded this spurious correlation despite it ranking last overall.

## 5. Lessons Learned and Future Pipeline Implications

Several important lessons emerged from this exploration, many of which connect directly to predictive performance outcomes. First, the moderate AUC-ROC of 0.68 achieved by the baseline Gradient Boosting Classifier (improved to 0.69 after grid search) revealed that the model had limited discriminative power—a performance ceiling that XAI helped explain rather than conceal. The SHAP attribution analysis showed that `prior_admissions` dominated predictions (mean $|\text{SHAP}| = 0.558$) while several other features contributed minimally, suggesting that feature engineering—such as admission history interaction terms—could materially improve predictive performance in future iterations. XAI did not merely describe the model; it diagnosed it.

Second, the detection of a non-zero attribution for `day_of_week` demonstrated that the model had learned a spurious correlation from training data, which would silently degrade generalization on future data where weekend coding patterns differ. Without XAI, this predictive performance risk would have been entirely invisible at model evaluation time. This underscores the need to treat XAI as a diagnostic complement to standard evaluation metrics, not a substitute.

Third, the comparison between the best GBM (AUC-ROC 0.69) and the MLP (AUC-ROC 0.66) highlighted a performance-interpretability tradeoff: the MLP was required for Integrated Gradients due to its smooth output surface, yet it performed slightly worse. Future pipelines should incorporate both model types in parallel—the best-performing model for prediction and a differentiable surrogate for IG-based attribution—so that neither predictive performance nor explanatory fidelity is sacrificed. Platform vendor lock-in, illustrated by the deprecation of Vertex Explainable AI, further reinforces the need for open-standard tools such as SHAP (Lundberg & Lee, 2017) that remain portable and independent of provider roadmaps.

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